This is the best method of sterilizing dry glass ware such as test tubes, petri dishes, flasks, pipettes and instruments such as forceps, scalpels and scissors. It is also used to sterilize some pharmaceutical products such as liquid paraffin, dusting powder, fats and grease.

Quality control of dry heat sterilization:

The spores of a nontoxigenic strain of *Clostridium tetani* are used to test the efficiency of dry heat sterlization.

4.4.2 Sterilization by Moist Heat

Moist heat kills microorganisms primarily by the coagulation of proteins (denaturation), which is caused by breakage of the hydrogen bonds that hold the proteins in three dimensional structure.

There are three methods employed in moist heat sterilization.

- Temperature below 100°C.
- Temperature at 100°C.
- Temperature above 100°C.

a) Temperature below 100°C:

Pasteurization

The process of heating a liquid food or beverage either at 62.8°C for 30 minutes or 72°C for 15 seconds to enhance their shelf life and destroy harmful microorganisms. It should be noted that pasteurization process kills only vegetative cells but not the spores. Pasteurization named in honour of its developer Louis Pasteur. Table 4.1 gives comparison between Sterilization and Pasteurization.



Raw milk can harbour dangerous microorganisms, such as *Salmonella, Escherichia coli* and *Listeria*, that can pose serious health risk, and children are particularly susceptible to the potential infection of unpasteurized or raw milk

Pasteurization can be done in the following methods,

- **Low Temperature Holding Method (LTH)**
  In this method milk, beer and fruit juices are maintained at 62.8°C for 30 minutes.
- **High Temperature Short Time Method (HTST)**
  Products are held at 72°C for 15 seconds.
- **Ultra High Temperature (UHT)**

Table 4.1: Comparison between Sterilization and Pasteurization

Sterilization	Pasteurization
Sterilized products have a longer shelf life	Pasteurized products have shorter shelf life
Discovered by Nicolas Appert	Discovered by Louis Pasteur
Eliminates all forms of microorganisms	Eliminates pathogenic microorganisms only
Can be accomplished in many ways	Can be accomplished with heat
Applied in food industry, medical, surgery and packaging	Mainly applied in food industry

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